Initiation to Statistics in Proteomics Analysis

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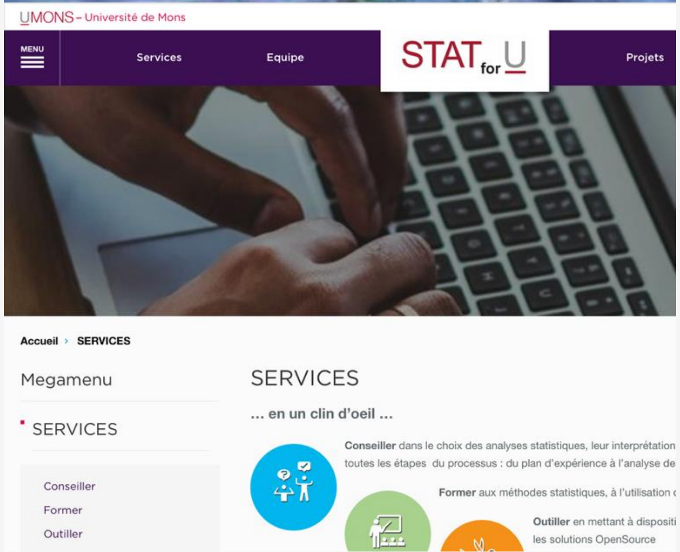


Why Statistics Matter in Proteomics

Proteomics generates massive datasets requiring rigorous statistical frameworks to extract biological meaning.

Modern mass spectrometry identifies thousands of proteins across multiple samples

Biological variability and technical noise create uncertainty in measurements

Statistical methods distinguish true biological signals from random variation

Proper analysis prevents false discoveries and missed findings

~10K

Proteins per sample

95%

Noise Reduction

1

Reliable Conclusion

What is Probability?

Probability quantifies uncertainty as a number between 0 and 1 representing the likelihood of an event.

P(Event) = Favorable Outcomes / Total Outcomes

Classical	Based on theoretical models (e.g., fair coin: P(Heads) = 0.5)
Frequentist	Long-run frequency of events in repeated experiments
In Proteomics	Probability that a protein is truly differentially expressed given observed data

$$0 \leq P \leq 1$$

The Axiom of Probability

Measuring Probability Experimentally

Experimental Formula

P(E) ≈ n(E) / n(Total)

Law of Large Numbers

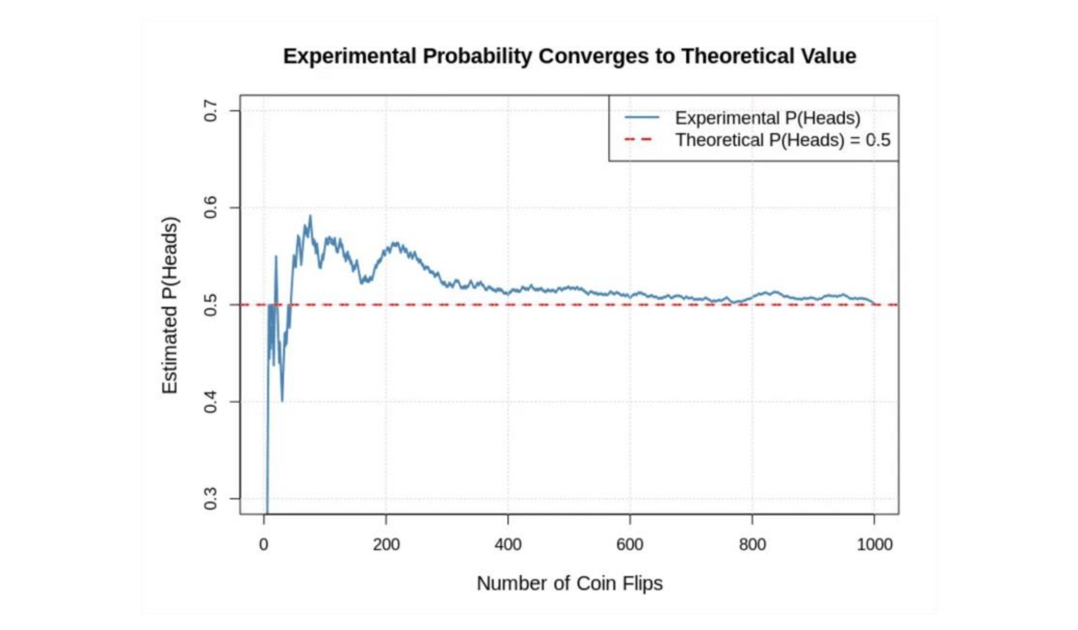
As sample size increases, experimental probability converges to the theoretical probability.

Example

Flip a coin 100 times, observe 53 heads → P(Heads) ≈ 0.53.

In Proteomics

Estimate detection probability by counting occurrences across technical replicates. Larger sample sizes yield more reliable estimates.



Simulation of 1000 coin flips showing convergence to P = 0.5

The foundation for statistical testing, characterized by Mean (μ) and Standard Deviation (σ).

68-95-99.7 Rule:

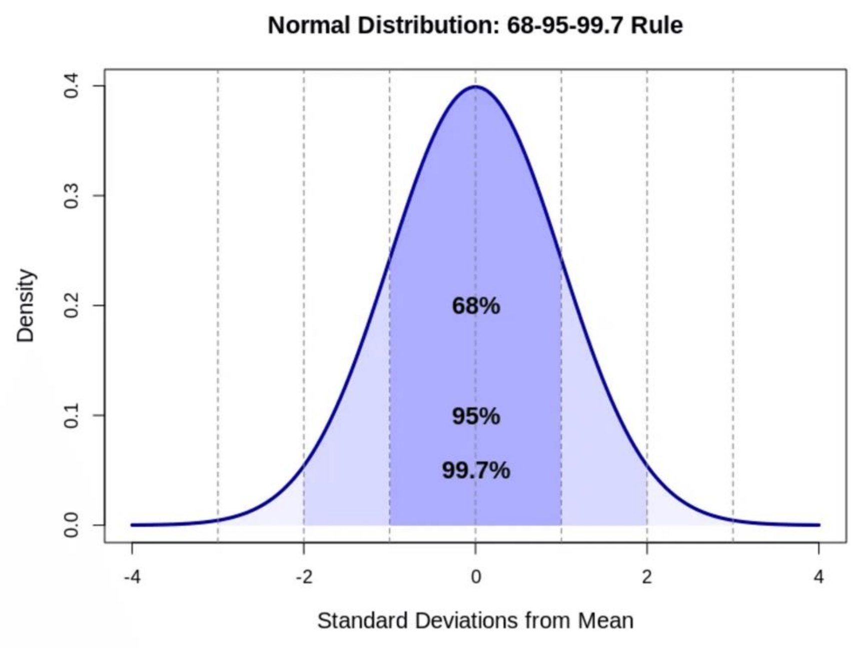
Data falls within 1σ, 2σ, and 3σ respectively.

Many biological measurements, including protein abundance, approximate this distribution (or the log of the measurements).

Essential for modeling noise and calculating probabilities in proteomics.

Central Limit Theorem

Sample means follow a normal distribution regardless of the underlying data distribution, enabling robust statistical inference.



Statistical Hypothesis Testing Framework

H_0

Null Hypothesis

The default assumption of no effect or no difference.

Example: Protein abundance is equal between conditions.

H_1

Alternative Hypothesis

The effect we want to detect.

Example: Protein abundance differs between conditions.

01 Collect data and calculate a **test statistic** that measures evidence against H_0.

02 Compare the statistic to a known distribution (e.g., Normal, t-distribution).

03 If evidence is strong enough ($p < \alpha$), **reject H_0** in favor of H_1.

Key Principle

We never "prove" H_1 is true, we only reject or fail to reject H_0.

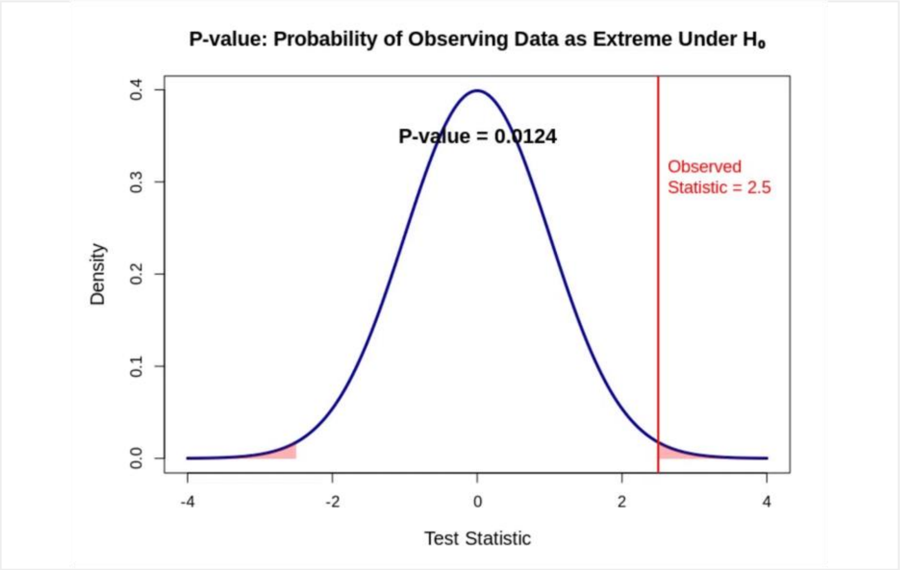
P-value = P(Data | H_0)

The probability of observing data as extreme or more extreme than what was observed, assuming the null hypothesis is true.

Common Misconception

It is NOT the probability that the null hypothesis is true.

- → **Threshold:** $p < 0.05$ (5% significance level) is standard.
- → **Decision:** Small p-value suggests data is unlikely under H_0 → Reject H_0.
- → **Example:** $p = 0.03$ means there is a 3% chance of seeing this difference by random noise alone.



Visualizing the p-value as the tail area under the null distribution curve.

Type I and Type II Errors

Type I Error (α)

False Positive

Rejecting H_0 when it is actually true.

> *"Claiming a protein is differentially expressed when it's not."*

Type II Error (β)

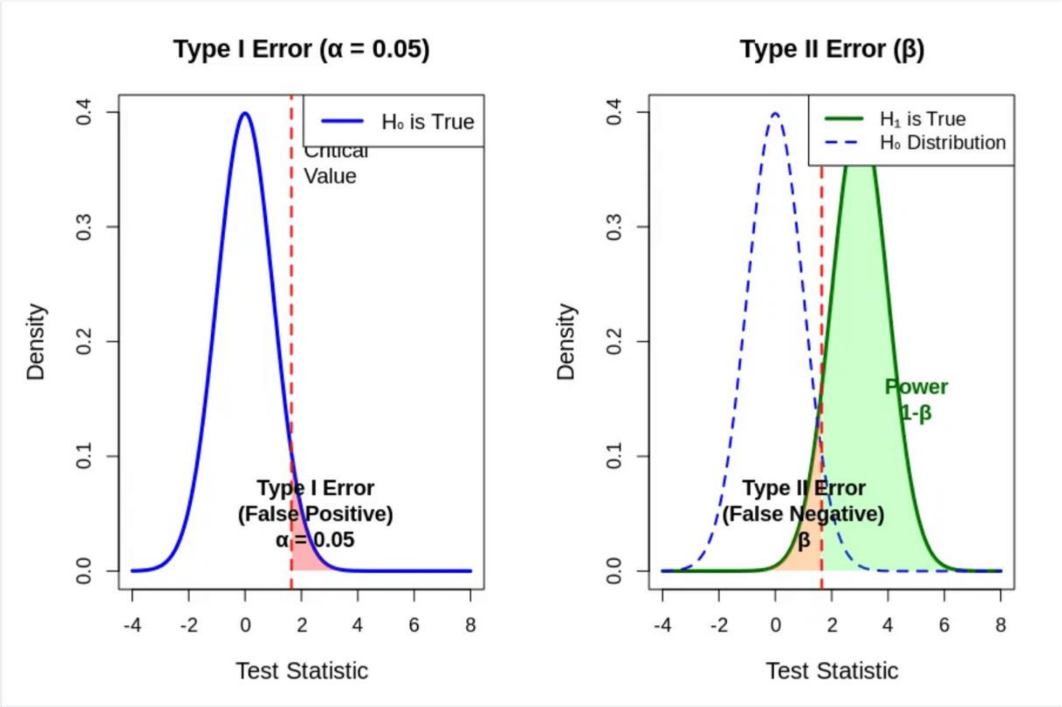
False Negative

Failing to reject H_0 when H_1 is true.

> *"Missing a truly differentially expressed protein."*

The Trade-off

Reducing Type I errors (stricter α) inevitably increases Type II errors (lower power).



Visualizing the overlap between Null (H_0) and Alternative (H_1) distributions

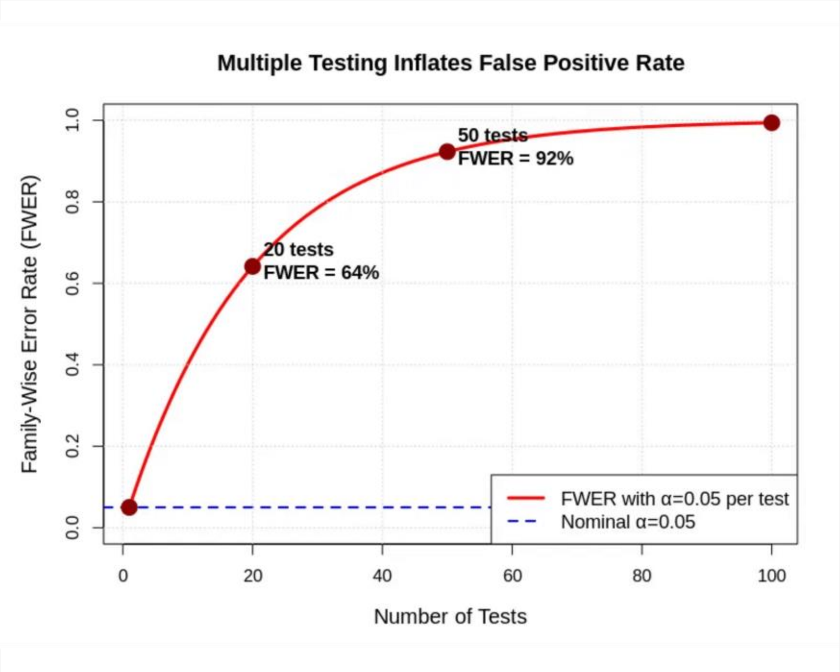
Testing thousands of proteins simultaneously inflates false positive rates dramatically without correction.

The Numbers Game

10,000 Proteins
× 0.05 (α)
= 500 False Positives

Family-Wise Error Rate (FWER)

The probability of making at least one Type I error. Without correction, FWER approaches 100% as the number of tests increases.



Even with just 20 tests, the probability of at least one false positive exceeds 60%.

Bonferroni Correction

The Method

$$\alpha\_adj = \alpha / n$$

Example: 10,000 proteins, $\alpha = 0.05$

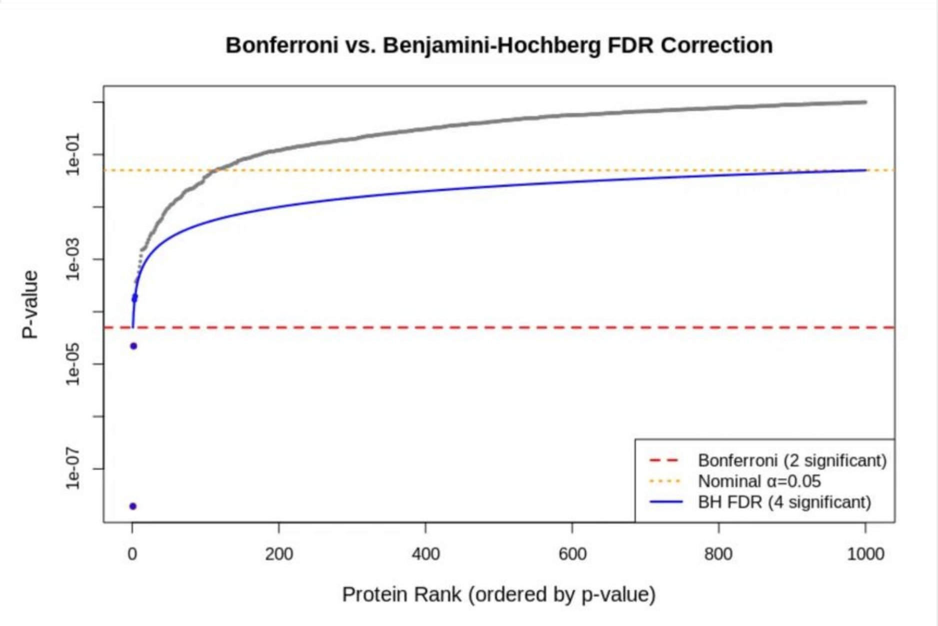
Threshold = 0.000005

Advantage

Simple to implement and strictly guarantees Family-Wise Error Rate (FWER) ≤ α.

Disadvantage

Extremely conservative. Drastically reduces statistical power and increases Type II errors (missed discoveries).



Comparison showing the strict Bonferroni threshold (red dashed line) vs the more adaptive FDR threshold.

False Discovery Rate (FDR)

Definition

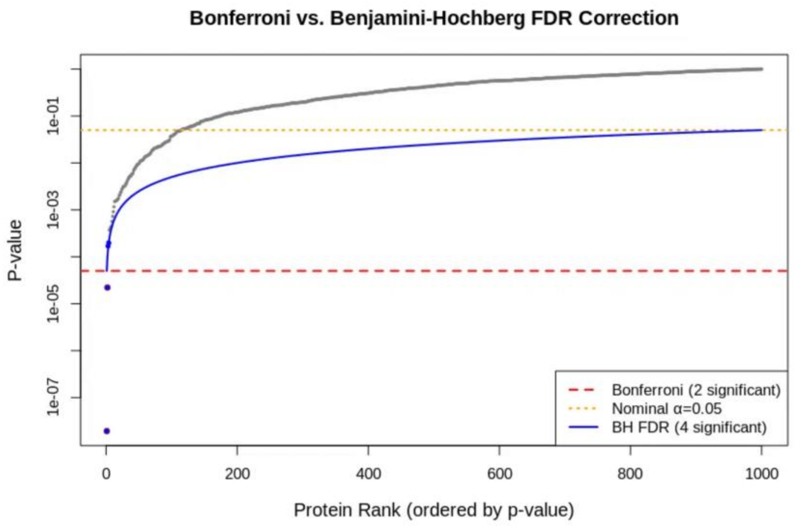
The expected proportion of false positives among all rejected hypotheses (discoveries).

Benjamini-Hochberg (BH) Procedure

A step-up procedure that controls FDR. It is less conservative than Bonferroni, offering higher statistical power.

Proteomics Standard

FDR < 0.05 implies we accept that 5% of our "significant" proteins may be noise in exchange for finding more true biological signals.



FDR (Blue) adapts the threshold to allow more discoveries than Bonferroni (Red).

Confusion Matrix in Classification

Precision (PPV)

TP / (TP + FP)

Proportion of predicted positives that are actually correct.

Recall (Sensitivity)

TP / (TP + FN)

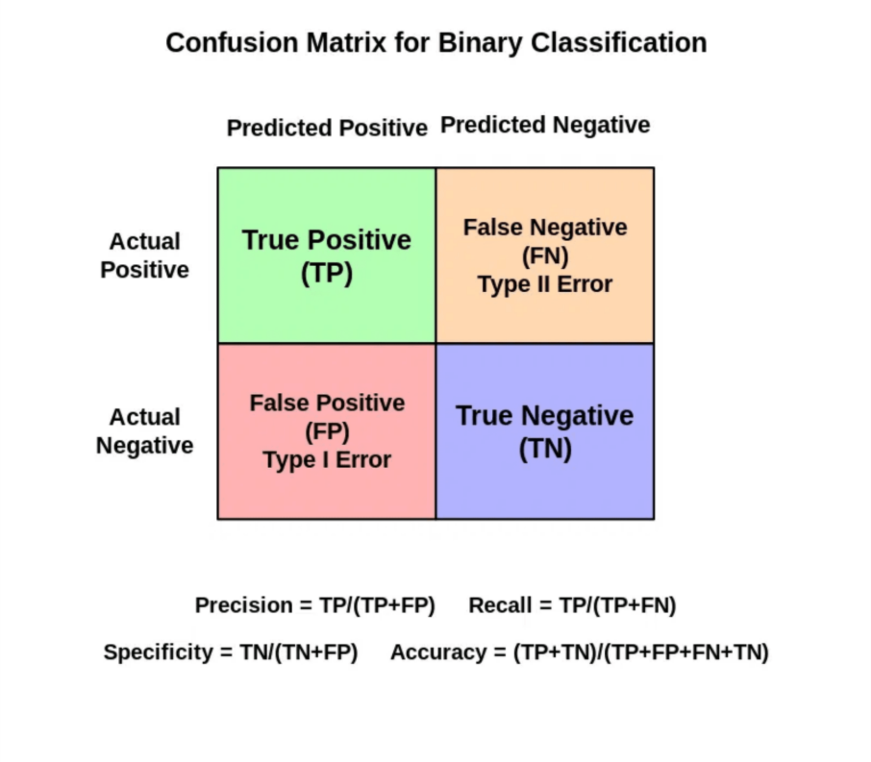
Proportion of actual positives correctly identified.

Specificity

TN / (TN + FP)

Proportion of actual negatives correctly identified.

Trade-off: Increasing recall often decreases precision.



Precision-Recall Trade-off

Strict Threshold

High Precision, Low Recall.
Few false positives, but many missed discoveries (False Negatives).

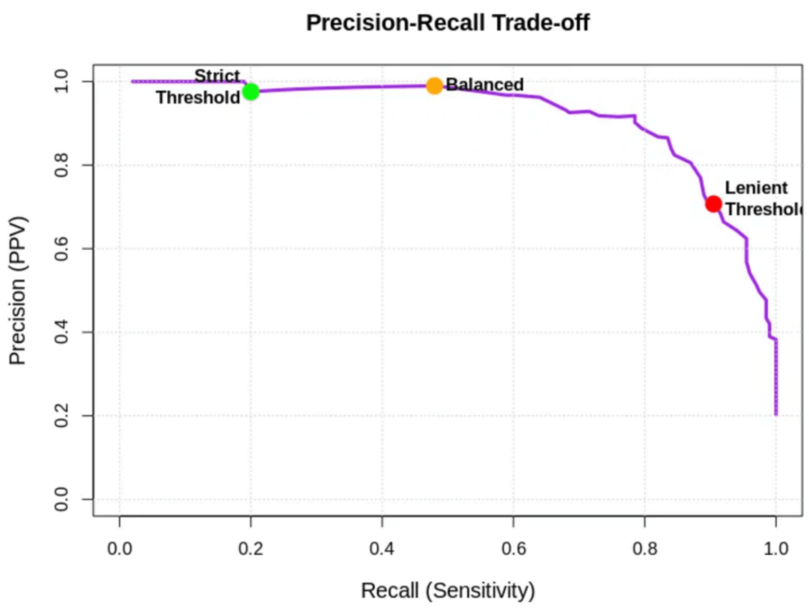
Lenient Threshold

Low Precision, High Recall.
Captures most true signals, but includes many false positives.

Strategic Application

Discovery Phase:
Favor Recall (Lenient) to cast a wide net.

Validation Phase:
Favor Precision (Strict) to ensure reliability.



The inverse relationship between Precision (PPV) and Recall (Sensitivity)

Bayes' Theorem Fundamentals

$$P(target|positive) = \frac{P(positive|target)\,P(target)}{P(positive)}$$

The Logic of Inference

Target here is a protein that is over- or under-expressed

$P(target|positive)$ **Posterior / Positive Predictive Value (PPV)**

Probability of Hypothesis given Data (The Goal).

$P(positive|target)$ **Likelihood / sensitivity of the target detection**

Probability of Data given Hypothesis (The Evidence).

$P(target)$ **Prior / prevalence = fraction of protein whose expression change**

Initial belief about Hypothesis before seeing data.

$P(positive)$ **Marginal / fraction of proteins change detected (correct or not)**

Total probability of observing the Data.

Core Concept

Bayes' theorem provides a mathematical way to update our prior beliefs with new experimental evidence.

Bayes' Theorem and False Positives

Low prevalence of true positives dramatically increases false positive rates, even with accurate tests.

The 1% Prevalence Problem

Sensitivity: **95%**

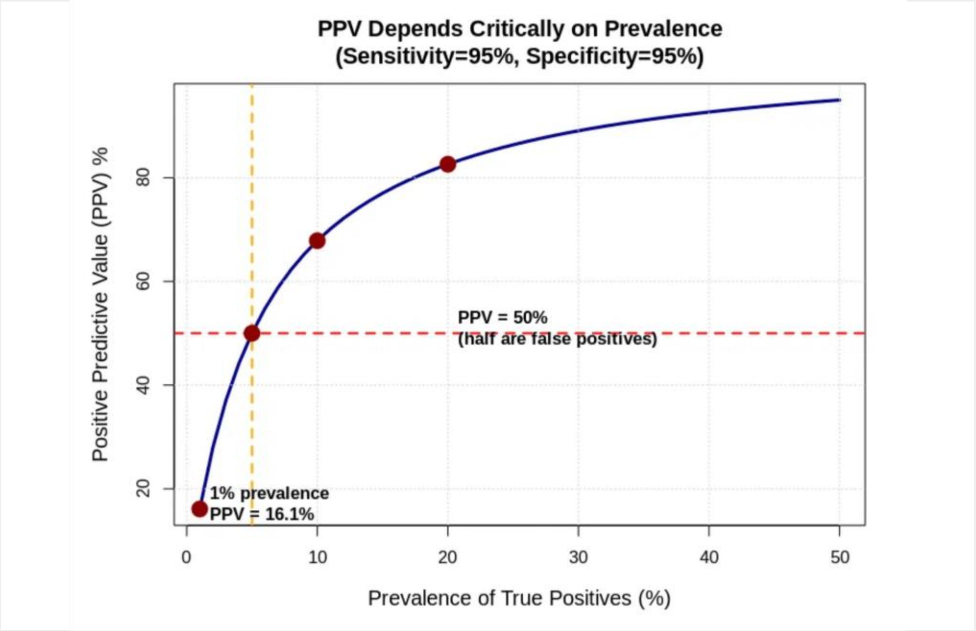
Specificity: **95%**

Prevalence: **1%**

Positive Predictive Value (PPV) **16%**

84% False Positives!

"In proteomics, if only 1% of proteins are truly changing, standard p-value thresholds will flood your results with noise. You need stricter controls (low FDR)."



As prevalence drops (x-axis), the reliability of a positive result (y-axis) plummets.

Differential Expression Analysis in Proteomics

01. Quantify

Measure protein abundance.

02. Normalize

Remove technical bias.

03. Statistical Testing

Compute p-values per protein.

04. Correction

Adjust p-values (FDR).

Common Statistical Tests

- **t-test:** 2-group comparison.
- **ANOVA:** >2 groups.
- **limma:** Linear models for complex designs.

Key Assumptions

Data ~ normal (often log-transformed).

Independent samples.

Comparable group variances.

The Goal

List proteins with significant fold changes and adjusted p-values to identify candidates.

Practical Example - Data Preparation in R

Essential Packages

limma	Linear Models
DEqMS	Variance Modeling
ggplot2	Visualization

Data Structure

Expression Matrix:

Rows = Proteins

Columns = Samples (Replicates)

Workflow

1. Import Data
2. Filter Missing Values
3. Normalize (Quantile/Median)
4. Log Transformation

```r
# Load required libraries
library(limma)
library(DEqMS)

# 1. Load Data (Expression Matrix)
# Rows = Proteins, Columns = Samples
data <-read.csv("protein_abundance.csv", row.names=1)
# 2. Quality Control & Filtering

# Keep proteins with valid data in at least 3 samples
valid_rows <-rowSums(!is.na(data)) >=3
filtered_data <- data[valid_rows, ]

# 3. Normalization (Quantile)
# Corrects for technical variation between samples
norm_data <-normalizeQuantiles(as.matrix(filtered_data))

# 4. Log Transformation
log_data <-log2(norm_data +1)
```

Practical Example - Statistical Testing in R

The limma Package

Originally designed for microarrays, **limma** (Linear Models for Microarray Data) is the gold standard for proteomics differential expression.

Empirical Bayes Moderation

Borrows information across all proteins to stabilize variance estimates. Critical for small-sample proteomics (n = 3 to 6).

Why it works

Prevents proteins with artificially low variance from appearing significant by chance.

```r
# Fit linear model and contrasts, then apply eBayes
Fit <- lmFit(expression_data, design)
contrast_matrix <- makeContrasts(TreatmentVsControl=Treatment-Control, levels=design)
fit2 <- contrasts.fit(fit, contrast_matrix)
fit2 <- eBayes(fit2)
results <- topTable(fit2, number=Inf)
```

lmFit()

Fits a linear model for each protein.

eBayes()

Computes moderated t-statistics and p-values.

Practical Example - Multiple Testing Correction in R

The p.adjust() Function

R's built-in function to adjust raw p-values using various methods. For proteomics, **"BH" (Benjamini-Hochberg)** is the standard for FDR control.

The Reality Check

Protein A	p = 0.01 → **adj.p = 0.15**
Protein B	p = 0.0001 → **adj.p = 0.03**

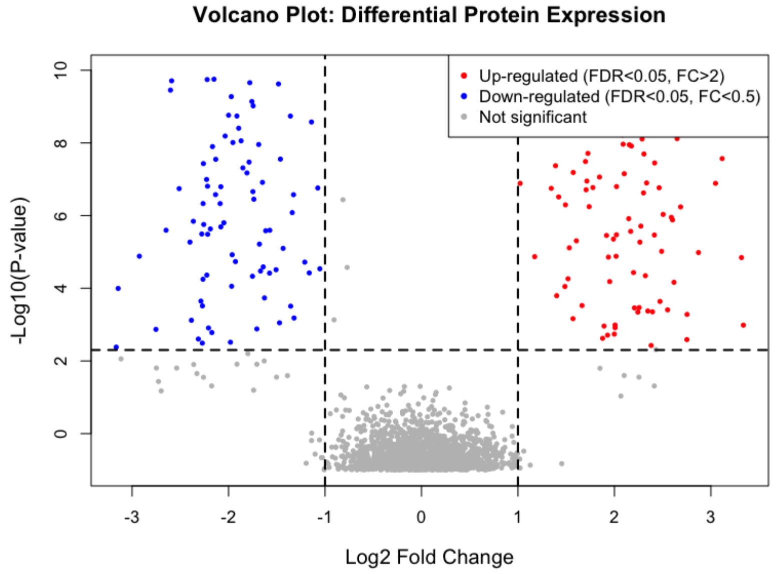
Protein A is no longer significant after correction!

```
# 1. Extract raw p-values from results
raw_pvalues <- results$P.Value

# 2. Apply Benjamini-Hochberg (FDR) correction
results$adj.P.Val <- p.adjust(raw_pvalues, method = "BH")

# 3. Filter for significant discoveries (FDR < 0.05)
significant_proteins <- subset(results, adj.P.Val < 0.05)

# 4. Check how many proteins passed
print(nrow(significant_proteins))
```

Practical Example - Results Visualization in R

Generating a Volcano Plot

```r
library(ggplot2)

# Create Volcano Plot
ggplot(results, aes(x = logFC, y = -log10(adj.P.Val))) +
  geom_point(aes(color = Significant)) +
  theme_minimal() +
  labs(
    title = "Volcano Plot: Differential Protein Expression",
    x = "Log2 Fold Change",
    y = "-Log10(P-value)"
  ) +
  geom_hline(yintercept = -log10(0.05), linetype = "dashed") +
  geom_vline(xintercept = c(-1, 1), linetype = "dashed")
```



X-Axis (Log2 Fold Change)

Magnitude of change. Positive = Upregulated, Negative = Downregulated.

Y-Axis (-Log10 P-value)

Statistical significance. Higher points are more significant.

Summary and Best Practices

01 Power in Numbers

Biological replicates are essential. $n = 3$ is the absolute minimum; $n \geq 5$ is recommended for robust statistics.

02 Choose the Right Test

Use **limma** or similar moderated tests for small sample sizes to stabilize variance estimates.

03 Correct for Multiple Testing

Always apply **FDR correction** (Benjamini-Hochberg). A raw p-value < 0.05 is rarely sufficient in proteomics.

04 Validate Findings

Statistical significance ≠ Biological relevance. Validate key hits with orthogonal methods (e.g., repetition, Western Blot, biological validation).

Core Philosophy

"Statistics is a filter to remove noise, not a machine that generates truth."

Use statistical tools to prioritize candidates, but let biological understanding guide your final conclusions.

Resources and Further Learning

Essential Reading

FDR & Benjamini-Hochberg

A Tutorial on False Discovery Control

https://www.stat.cmu.edu/~genovese/talks/hannover1-04.pdf

Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing (1995)

https://rss.onlinelibrary.wiley.com/doi/10.1111/j.2517-6161.1995.tb02031.x

Software & Tools

Platform

Bioconductor

Open-source software project for high-throughput genomic/proteomic data https://www.bioconductor.org

R Package

limma

Linear models for microarrays and omics data https://www.bioconductor.org/packages/release/bioc/html/limma.html

Proteomics with Open Source Software

Proteomics Analysis with R

Using R for Proteomics Data Analysis (nov. 2025)

https://www.bioconductor.org/packages/release/data/experiment/vignettes/RforProteomics/inst/doc/RforProteomics.html

R for Mass Spectrometry (2026)

https://www.rformassspectrometry.org

Proteomics Analysis with Python

Pyteomics

https://pyteomics.readthedocs.io/en/latest/

This Presentation & Associated R Scripts

https://filedn.com/lzGVgfOGxb6mHFQcRn9ueUb/bepa2026/bepa2026_stats.pdf
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